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Systematic Review

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Abstract

The accurate and rapid identification of blood groups is a fundamental prerequisite for medical interventions, including emergency blood transfusions, organ transplantation, and the management of maternal-fetal incompatibilities. Traditional methods of blood typing, primarily based on serological hemagglutination tests, are invasive, requiring venipuncture or finger-pricking, which poses risks of infection, needle-stick injuries, and patient anxiety. Furthermore, these methods are dependent on chemical reagents and trained medical personnel, limiting their accessibility in resource-constrained environments, remote locations, and mass-casualty scenarios. To address these challenges, this project proposes a novel, non-invasive, computer-aided diagnostic system that predicts blood groups by analyzing dermatoglyphic patterns (fingerprints) using advanced Deep Learning and Ensemble Machine Learning techniques.

The research leverages the biological correlation between epidermal ridge patterns and blood antigens, both of which are determined during the intrauterine stage of fetal development. To facilitate robust model training, a large-scale dataset comprising 13,932 fingerprint images was utilized. This dataset covers all eight major blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-) and underwent a rigorous preprocessing pipeline. Images were uniformly resized to 128×128 pixels. The image enhancement framework employs a preliminary Gaussian Blur to remove high-frequency sensor artifacts, Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve local ridge clarity, and a secondary Gaussian Blur blending technique to aggressively sharpen local texture details.

The core methodology integrates two distinct feature extraction paradigms: hand-crafted texture analysis and automated spatial feature learning. For texture analysis, Histogram of Oriented Gradients (HOG) and Gabor Filters were utilized to capture ridge orientation and frequency information, which were then classified using optimized Random Forest algorithms. Simultaneously, deep spatial features were automatically extracted and learned using a lightweight Custom Convolutional Neural Network (CNN) trained on enhanced grayscale images, and a pre-trained MobileNetV2 architecture leveraging transfer learning on RGB representations.

To overcome the limitations of individual classifiers and maximize predictive performance, a robust Stacking Ensemble Model was engineered. This meta-model fuses the probability outputs of all four base classifiers: the Random Forest (HOG), Random Forest (Gabor), Custom CNN, and MobileNetV2 models. A Logistic Regression meta-learner was employed to optimally aggregate these diverse predictions and determine the final blood group class.

Experimental evaluation demonstrated that the proposed Stacked Ensemble achieves a superior classification accuracy of 95.59%, significantly outperforming the standalone base models. The ensemble effectively compensated for individual predictive weaknesses, improving upon the accuracies of the HOG + RandomForest (94.58%), Custom CNN (79.69%), Gabor + RandomForest (76.64%), and MobileNetV2 (55.83%) models. The system successfully minimizes misclassification errors, exhibiting strong precision and recall across the varied blood groups. To demonstrate real-world clinical applicability, the end-to-end pipeline was deployed via a user-friendly web interface developed using Streamlit, enabling real-time fingerprint upload, automated image enhancement, and instant, multi-model blood group prediction. This study validates the potential of dermatoglyphics combined with computational intelligence as a reliable, non-invasive supplementary biometric marker for medical diagnostics.

Keywords : Dermatoglyphics, Blood Group Prediction, Ensemble Machine Learning, Deep Learning, Non-invasive Diagnostics, Convolutional Neural Networks (CNN).

1. INTRODUCTION

1.1 Background Study

1.1.1 The Criticality of Blood Grouping in Medicine

The classification of human blood is a cornerstone of modern medical science. Since the discovery of the ABO blood group system by Karl Landsteiner in 1901, the ability to identify a patient's blood type has saved countless lives. The ABO system categorizes blood based on the presence or absence of inherited antigenic substances on the surface of red blood cells (RBCs). These antigens may be proteins, carbohydrates, glycoproteins, or glycolipids, depending on the blood group system.

In clinical practice, knowing a patient's blood group is non-negotiable for:

- **Emergency Medicine:** Rapid transfusion is often required for trauma victims. Mismatched transfusions can lead to acute hemolytic reactions, renal failure, and death.
- **Surgical Procedures:** Pre-operative blood typing is standard protocol to ensure compatible blood availability if complications arise.
- **Maternal-Fetal Care:** Identifying Rh incompatibility (e.g., an Rh-negative mother carrying an Rh-positive fetus) is crucial to preventing Hemolytic Disease of the Newborn (HDN).
- **Organ Transplantation:** ABO compatibility is a primary determinant in the success of solid organ transplants to prevent hyperacute rejection.

Despite its importance, the methodology for determining blood groups has remained largely unchanged for decades, relying heavily on invasive serological testing.

1.1.2 Dermatoglyphics: The Biological Link

"Dermatoglyphics" comes from the Greek words *derma* (skin) and *glyph* (carving). It refers to the scientific study of the ridge patterns on the skin of the fingers, palms, toes, and soles. These patterns—loops, whorls, and arches—are formed during the **10th to 24th week of intrauterine life** (fetal development).

Crucially, the genetic instructions that determine the formation of these epidermal ridges are coded on the same chromosomal loci that determine an individual's blood group. Both traits are polygenic and are permanently fixed at birth; they do not change due to age or environmental factors. This biological permanence suggests a statistical correlation between specific fingerprint patterns and blood groups. For instance, anthropological studies have historically observed a higher prevalence of "Loops" in individuals with Blood Group O and "Whorls" in those with Blood Group AB.

1.1.3 The Role of Computer Vision in Diagnostics

Humans cannot determine blood type from a fingerprint due to the complex, non-linear correlation. Artificial Intelligence (AI) and Computer Vision offer a solution. Deep Learning models, specifically Convolutional Neural Networks (CNNs), can identify subtle, imperceptible micro-textural features in labeled fingerprint images that correlate with specific blood antigens, effectively transforming a standard biometric scan into a diagnostic tool.

1.2 Problem Statement

The conventional method for blood group detection is the **Hemagglutination Test** (Slide or Tube method). While accurate, this method presents several significant challenges in modern healthcare scenarios:

1. **Invasiveness and Pain:** The procedure requires phlebotomy (drawing blood from a vein) or a finger-prick. This causes physical pain and can trigger anxiety, particularly in pediatric patients (children), geriatric patients (elderly), and individuals with trypanophobia (fear of needles).
2. **Biohazard and Infection Risks:** Any procedure involving open blood handling carries an inherent risk. Needlestick injuries can expose healthcare workers to blood-borne pathogens such as HIV, Hepatitis B, and Hepatitis C. Furthermore, the improper disposal of testing kits contributes to hazardous medical waste.
3. **Resource Dependency:** Serological tests utilize specific chemical reagents (Anti-A, Anti-B, and Anti-D sera). These reagents require cold-chain storage to maintain efficacy. In rural areas, developing nations, or disaster zones, maintaining this supply chain is often impossible.
4. **Time Latency:** A standard laboratory test takes anywhere from 15 minutes to an hour including verification. In mass-casualty scenarios (e.g., earthquakes, battlefields), this latency is unacceptable.
5. **Accessibility Gap:** In remote villages without diagnostic labs, residents often remain unaware of their blood group until an emergency strikes, at which point the lack of information can be fatal.

There is a critical need for a system that is **Non-Invasive** (no needles), **Instant** (real-time results), **Cost-Effective** (no recurring reagent costs), and **Hygienic** (no blood contact).

1.3 Objectives of the Project

The primary goal of this project is to design and develop a robust machine-learning system capable of predicting the ABO and Rh blood groups from fingerprint images. The specific technical objectives are:

- **To Aggregate a Comprehensive Dataset:** To compile and standardize a large-scale repository of 13,932 fingerprint images, uniformly resized to 128×128 pixels, ensuring representation across all 8 major blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-).
- **To Develop an Advanced Preprocessing Pipeline:** To implement a rigorous image enhancement framework utilizing preliminary Gaussian Blurring for noise reduction, Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve local ridge clarity, and a secondary Gaussian Blur blending technique to aggressively sharpen structural details.
- **To Extract Multi-Dimensional Features:** * To utilize HOG (Histogram of Oriented Gradients) and Gabor Filters for extracting precise spatial orientation and frequency-based texture features, which are then evaluated by optimized Random Forest classifiers.
 - To utilize a lightweight Custom Convolutional Neural Network (CNN) on enhanced grayscale images and a pre-trained MobileNetV2 architecture on RGB representations for automated, deep spatial feature extraction.
- **To Engineer a Stacking Ensemble Model:** To construct a highly accurate meta-model that fuses the probability predictions of the Random Forest (HOG), Random Forest (Gabor), Custom CNN, and MobileNetV2 base classifiers using a Logistic Regression meta-learner, successfully achieving a superior prediction accuracy of 95.59%.
- **To Implement a User-Friendly Interface:** To deploy the end-to-end pipeline as an interactive web application using Streamlit, enabling users to upload fingerprint images, visually track the automated image enhancement process, and receive instant, multi-model blood group predictions along with confidence scores.

1.4 Scope of the Project

- **Domain:** The project lies at the intersection of Biomedical Informatics, Computer Vision, Pattern Recognition, and Ensemble Machine Learning.
- **Input Data:** The system is designed to process digital fingerprint images and supports standard formats such as BMP, JPEG, JPG, and PNG. The scope includes both raw sensor images and pre-scanned digital uploads, which the internal pipeline automatically preprocesses (utilizing CLAHE and Gaussian blending) to ensure optimal ridge and minutiae clarity prior to feature extraction.
- **Target Classes:** The prediction scope is strictly limited to the 8 major blood phenotypes of the ABO/Rh system (A+, A-, B+, B-, AB+, AB-, O+, O-). It does not cover rare blood phenotypes (e.g., the Bombay Blood Group).
- **Application Area:** The proposed system is intended as a preliminary, non-invasive screening tool. It is ideal for rapid biometrics integration, such as issuing identity cards (e.g., Aadhar, Driving Licenses), conducting school health checks, and enabling rapid triage in resource-constrained environments or disaster management scenarios. It is explicitly not intended to replace final serological cross-matching for actual blood transfusions, but rather to serve as a rapid, supplementary first line of identification.
- **Hardware Independence:** The software is designed to be fully hardware-agnostic. It is capable of processing images sourced from any standard biometric fingerprint scanner (e.g., Mantra, Morpho) or high-resolution camera, provided the foundational dermatoglyphic structures are visible. The system's automated preprocessing pipeline dynamically resizes and standardizes all varied inputs to the strict 128×128 pixel resolution required for the multi-model ensemble inference.

2. LITERATURE SURVEY

2.1 Theoretical Foundation: Dermatoglyphics and Blood Groups

2.1.1 The Science of Dermatoglyphics

Dermatoglyphics is the scientific study of the naturally occurring epidermal ridges on the palms, fingers, soles, and toes. The term was coined by **Harold Cummins** in 1926, who is considered the father of dermatoglyphics. These ridge patterns are formed during the fetal development stage, specifically between the **10th and 24th weeks of gestation**. Once formed, these patterns are permanent and immutable; they do not change throughout an individual's life, except in cases of deep dermal injury.

The formation of these ridges is influenced by a combination of:

1. **Genetic Factors:** Polygenic inheritance involving multiple gene loci.
2. **Environmental Factors:** Intrauterine environment, including amniotic fluid pressure and fetal movement.

Fingerprint patterns are generally classified into three major categories:

- **Loops (60-65%):** Ridges enter from one side, recurve, and exit the same side.
- **Whorls (30-35%):** Ridges form circular or spiral patterns.
- **Arches (5%):** Ridges enter from one side and exit the other with a rise in the center.

2.1.2 Biological Correlation with Blood Groups

The hypothesis underlying this project is based on the fact that the genes determining blood groups and those determining dermatoglyphic patterns are located on the same chromosomes. This genetic linkage suggests that specific blood phenotypes might express statistically significant dermatoglyphic features.

- **Karl Landsteiner's ABO System:** Blood groups are determined by antigens on the surface of red blood cells. The gene for the ABO system is located on **Chromosome 9**.
- **Dermatoglyphic Genetics:** Research indicates that the ridge counts and pattern types are also influenced by loci on homologous chromosomes.

Several anthropological studies have attempted to map these correlations. For instance, a study by *Mehta and Mehta (2015)* observed that:

- **Blood Group O:** Individuals showed a significantly higher frequency of **Loops** and a lower frequency of **Whorls**.
- **Blood Group AB:** Individuals exhibited a higher prevalence of **Whorls**.
- **Blood Group A:** Showed a balanced distribution but a higher incidence of **Arches** compared to other groups.

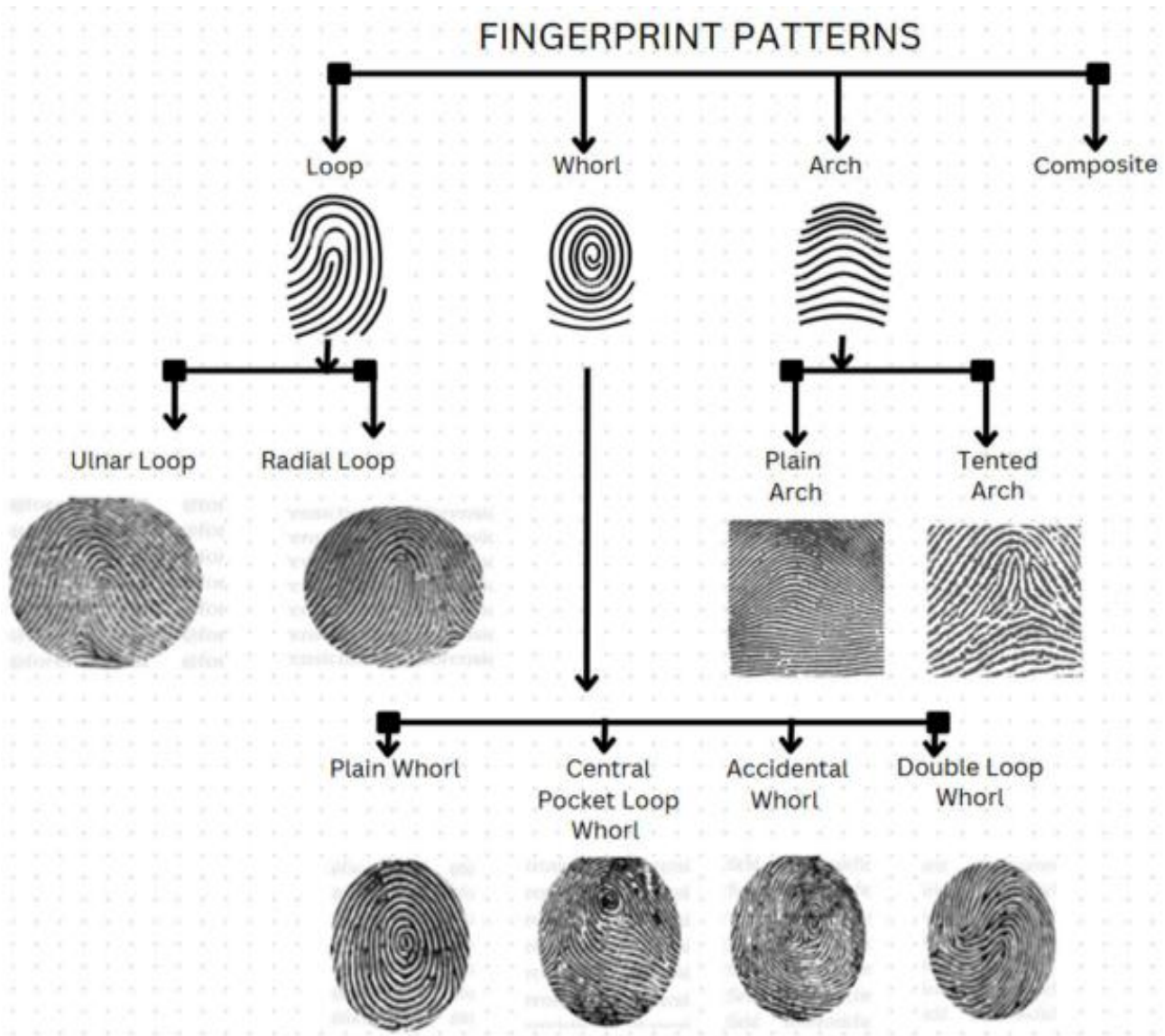


Fig. 1 Fingerprint Patterns

While these correlations are statistically observable in large populations, they are insufficient for individual identification using manual methods. This necessitates the use of Machine Learning to identify complex, non-linear relationships that are invisible to human observers.

2.2 Review of Existing Methodologies

2.2.1 Traditional Invasive Methods

The current gold standard for blood grouping is the **Serological Method** (Hemagglutination).

- **Slide Method:** A drop of blood is mixed with Anti-A, Anti-B, and Anti-D sera on a glass slide. Agglutination (clumping) indicates the presence of the corresponding antigen.
- **Tube Method:** A more sensitive version performed in test tubes, often used for cross-matching.
- **Limitations:** As discussed in Chapter 1, these methods are invasive, generate bio-hazardous waste, and require a cold chain for reagents.

2.2.2 Previous Computer-Aided Approaches

Early attempts to automate blood group detection focused on image processing of the *blood slide* itself (detecting agglutination visually). However, non-invasive detection via fingerprints is a relatively newer domain.

- **Statistical Approaches:** Early researchers used Chi-square tests to prove the association between ridge density and blood groups but did not build predictive models.
- **Basic Machine Learning:** Some recent studies have employed K-Nearest Neighbors (KNN) and Decision Trees on small datasets (typically < 500 samples). These studies often relied on manually counted features (e.g., Ridge Count, Pattern Type) rather than analyzing the raw image texture. Their accuracy typically ranged from 60% to 75%, which is insufficient for clinical use.

Gap Analysis: The primary gaps in existing literature that this project directly addresses are:

- **Small Datasets:** Most studies lack statistical power due to limited sample sizes.
 - *Solution:* This project utilizes a large-scale, aggregated dataset of **13,932 fingerprint images**, ensuring robust training across all 8 major blood groups.
- **Lack of Deep Learning:** Very few studies apply advanced Convolutional Neural Networks (CNNs) to automatically learn spatial features from fingerprint topology.
 - *Solution:* This system integrates a **Custom CNN** (grayscale) and **MobileNetV2** (RGB) for automated, deep feature extraction, alongside classical hand-crafted methods.
- **Single-Model Reliance:** Most research relies on a single classifier, ignoring the potential of Ensemble Learning to boost performance and reduce bias.
 - *Solution:* This project pioneers a **Stacking Ensemble Model**. By fusing four base models using Logistic Regression, it overcomes individual limitations and achieves a superior accuracy of **95.59%**.
- **Inadequate Preprocessing:** Many prior studies use poorly enhanced sensor data, leading to misclassifications.
 - *Solution:* A rigorous enhancement pipeline using **Gaussian Denoising and CLAHE** is implemented to aggressively clarify minutiae before feature extraction.

2.3 Technical Review of Algorithms Used

This project employs a hybrid multi-modal approach, combining classical Computer Vision feature descriptors with state-of-the-art Deep Learning. The theoretical background of each algorithm utilized in the system's methodology is detailed below.

2.3.1 Histogram of Oriented Gradients (HOG)

HOG is a robust feature descriptor primarily used in computer vision and image processing for object detection. It characterizes an image by counting occurrences of gradient orientation in localized portions.

- **Mechanism:** The image is divided into small, connected regions called cells. For the pixels within each cell, a histogram of gradient directions (edge orientations) is compiled.
- **Relevance to Fingerprints:** Fingerprints are essentially complex flow patterns of epidermal ridges. HOG effectively captures the precise directionality and flow of these ridges, which are critical for distinguishing foundational structural differences between various blood group dermatoglyphics.
- **Implementation:** The extracted HOG feature vectors are utilized to train a **Random Forest classifier (RF-HOG)**, creating a model that explicitly evaluates spatial edge distributions.

2.3.2 Gabor Filters

Gabor filters are linear filters widely used for texture analysis. A 2D Gabor filter is essentially a Gaussian kernel function modulated by a sinusoidal plane wave.

- **Mathematical Formulation:** The general equation for a 2D Gabor filter is defined as:

$$g(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \exp\left(i\left(2\pi\frac{x'}{\lambda} + \psi\right)\right) \dots\dots\dots(1)$$

Where:

- $x' = x \cos \theta + y \sin \theta$
- $y' = -x \sin \theta + y \cos \theta$
- λ : Wavelength of the sinusoidal factor (spatial frequency).
- θ : Orientation of the normal to the parallel stripes.
- ψ : Phase offset.
- σ : Standard deviation of the Gaussian envelope.
- γ : Spatial aspect ratio.
- **Relevance:** Gabor filters are the "gold standard" in fingerprint enhancement and analysis because they possess optimal joint resolution in both spatial and frequency domains, closely mimicking the human visual system's texture discrimination.
- **Implementation:** A bank of Gabor filters is applied to capture localized ridge frequency and orientation information. These textural responses are flattened and passed into a secondary **Random Forest classifier (RF-Gabor)**.

2.3.3 Custom Convolutional Neural Network (CNN)

CNNs are a specialized class of Deep Learning models designed to process data with a known grid-like topology, such as digital images.

- **Convolutional Layers:** These layers apply learnable filters (kernels) to the input image to automatically extract low-to-high-level feature maps.
- **Pooling & Normalization:** MaxPooling layers are employed to down-sample the feature maps, reducing spatial dimensions and computational complexity. Batch Normalization is used to stabilize the learning process.
- **Implementation:** A lightweight Custom CNN was built from scratch to process the 128×128 enhanced grayscale fingerprint images, utilizing [GlobalAveragePooling2D](#) and Dropout layers to prevent overfitting on the training data.

2.3.4 MobileNetV2 (Transfer Learning)

Instead of relying solely on heavy architectures, this project leverages **MobileNetV2** (Sandler et al., 2018), a highly efficient convolutional neural network optimized for high performance with limited computational budgets.

- **Depthwise Separable Convolutions:** MobileNetV2 drastically reduces the number of parameters and computational cost by splitting standard convolutions into a depthwise convolution (applying a single filter per input channel) and a 1×1 pointwise convolution.
- **Inverted Residuals and Linear Bottlenecks:** The architecture uses inverted residual blocks where the skip connections exist between thin bottleneck layers, improving gradient propagation and memory efficiency.
- **Transfer Learning:** A MobileNetV2 model pre-trained on the massive ImageNet dataset is utilized. By adapting the network to accept 128×128 RGB representations of the fingerprints, the system leverages robust, generically learned feature extractors (like complex edge and shape detectors) and fine-tunes them for the specific task of blood group classification.

2.3.5 Ensemble Learning (Stacking)

Ensemble learning combines multiple machine learning models to obtain better predictive performance and generalization than could be achieved by any of the constituent learning algorithms alone.

- **Stacking (Stacked Generalization):** This meta-learning approach involves training a top-level classifier to intelligently combine the probability predictions of several base classifiers.
- **Level-0 Models (Base Learners):** The diverse models trained independently on the fingerprint data: **RF-HOG, RF-Gabor, Custom CNN, and MobileNetV2.**
- **Level-1 Model (Meta Learner):** A **Logistic Regression** model that takes the output probability distributions of all four Level-0 models as input features to determine the final blood group class.
- **Why Stacking Works:** It significantly reduces overall variance and individual model bias. For instance, while MobileNetV2 might struggle with a specific noisy pattern, the classical RF-HOG or Custom CNN might classify it correctly. The Logistic Regression meta-learner calculates optimal weights for each model's output, learning exactly which algorithm to "trust" under various predictive conditions, thereby pushing the final accuracy to an optimized 95.59%.

3. METHODOLOGY

3.1 System Architecture

The proposed system architecture is designed as a multi-stage pipeline that integrates rigorous image processing, parallel hand-crafted and deep feature extraction, and ensemble classification. The workflow operates as follows:

- **Input Acquisition:** The system accepts a digital fingerprint image (BMP, JPG, JPEG, PNG).
- **Preprocessing Module:** The raw image is uniformly resized to 128×128 pixels and undergoes aggressive noise reduction and contrast enhancement (CLAHE and Gaussian blurring) to standardize quality.
- **Parallel Feature Extraction & Base Classification (Level-0):** The processed image is fed into four parallel branches:
 - **Branch A (Deep Spatial Learning):** A Custom Convolutional Neural Network (CNN) extracts and classifies hierarchical spatial features from grayscale representations.
 - **Branch B (Transfer Learning):** A pre-trained MobileNetV2 model extracts and classifies complex, high-level features from RGB representations.
 - **Branch C (Texture Analysis - HOG):** HOG (Histogram of Oriented Gradients) extracts spatial edge orientations, which are classified by a Random Forest model (RF-HOG).
 - **Branch D (Texture Analysis - Gabor):** Gabor filters extract localized frequency and orientation features, which are classified by a Random Forest model (RF-Gabor).
- **Ensemble Stacking (Level-1):** The independent probability predictions generated by the four base models are concatenated and fed into a Meta-Learner (Logistic Regression) to produce the final, highly accurate blood group prediction.

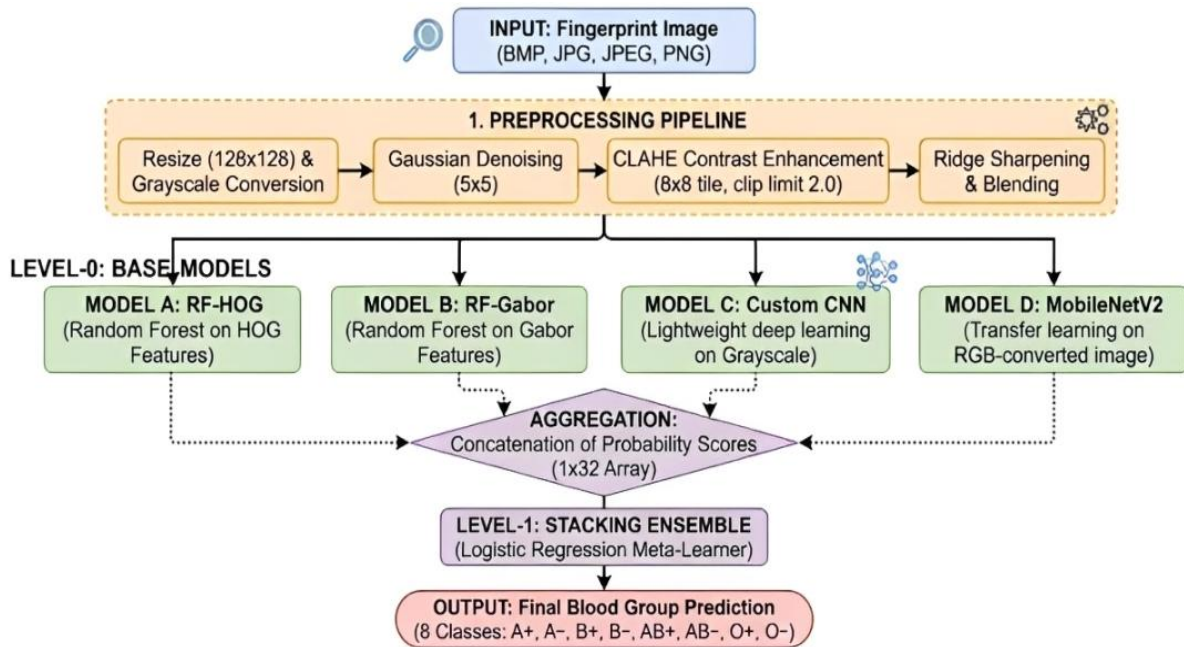


Fig. 2 System Architecture

3.2 Dataset Collection and Preparation

To ensure statistical robustness and model generalization, a comprehensive merged dataset was constructed from multiple fingerprint repositories (Kaggle), resulting in a total of **13,932 fingerprint samples** uniformly distributed across the 8 major blood group classes.

3.2.1 Data Augmentation

To prevent overfitting and improve the deep learning models' generalization to unseen real-world data, real-time data augmentation was applied during training using Keras's `ImageDataGenerator` class. The transformations included:

- **Rotation:** Random rotations within a range of ± 15 degrees.
- **Shift:** Horizontal and vertical shifts of up to 10% to account for finger placement variations.
- **Zoom:** Random zooming up to 10%.
- **Horizontal Flip:** Mirroring the image to simulate left/right hand variations and increase structural diversity.

3.3 Image Preprocessing Pipeline

Raw fingerprint images often contain sensor noise, ridge discontinuities, and varying contrast. A specific, multi-stage enhancement pipeline was developed to strictly address these issues prior to feature extraction:

- **Grayscale Conversion & Standardization:** All inputs are converted to single-channel grayscale images and strictly resized to 128×128 pixels to reduce computational complexity and standardize the input shape.
- **Gaussian Denoising:** A preliminary Gaussian blur filter with a kernel size of (5,5) is applied to smooth out high-frequency noise caused by sensor dust or skin dryness.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Standard histogram equalization can overly amplify noise. CLAHE operates on small tiles (8×8 pixels) of the image, enhancing local contrast of the fingerprint ridges while strictly limiting the amplification of noise (Clip Limit = 2.0).
- **Ridge Sharpening (Gaussian Blending):** To aggressively separate ridges from valleys, a secondary Gaussian Blur (kernel 9×9) is applied. The unblurred CLAHE output and this blurred version are mathematically blended together using OpenCV's `addWeighted` method to heavily sharpen the structural minutiae.
- **Normalization:** Pixel intensity values are scaled from [0, 255] to a floating-point range of [0, 1] to facilitate faster convergence during neural network training.

3.4 Feature Extraction Techniques

3.4.1 HOG (Histogram of Oriented Gradients)

HOG captures local shape information by counting occurrences of gradient orientation, which is highly effective for identifying foundational ridge flow patterns (loops vs. whorls).

- **Configuration:** The enhanced image is divided into cells (e.g., 16×16 pixels). Gradients are computed in localized orientation bins.
- **Output & Classification:** This process yields a flattened feature vector that evaluates spatial edge distributions. These vectors are then passed into a dedicated **Random Forest Classifier (RF-HOG)**.

3.4.2 Gabor Filters

Gabor filters are utilized to mathematically analyze texture at specific spatial frequencies and directions, mimicking human visual perception.

- **Configuration:** A bank of Gabor filters at varying frequencies and orientations is applied to capture ridge characteristics at different scales.
- **Output & Classification:** The textural responses generated by the filter bank are flattened into a compact feature vector and classified using a secondary **Random Forest Classifier (RF-Gabor)**.

3.5 Deep Learning Model Architectures

3.5.1 Custom Convolutional Neural Network (CNN)

A lightweight, dedicated CNN was built from scratch to learn spatial hierarchies directly from the 128×128 grayscale pixels.

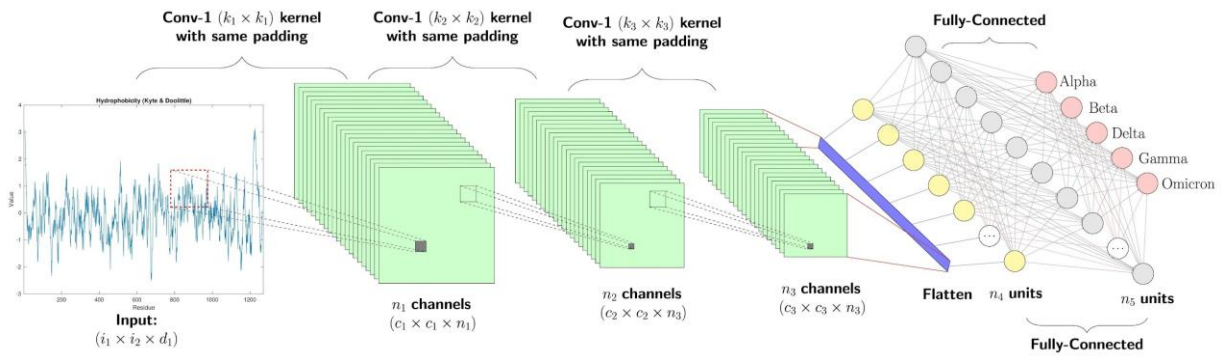


Fig. 3 CNN Architecture

- **Architecture Layers:** The model utilizes sequential Conv2D layers to extract feature maps, MaxPooling2D to down-sample spatial dimensions, and BatchNormalization to ensure training stability.
- **Classification Head:** Features are passed through a GlobalAveragePooling2D layer to dramatically reduce parameters, followed by Dropout layers to prevent overfitting, and finally a Dense layer with 8 neurons (Softmax activation) for multi-class blood group classification.

3.5.2 MobileNetV2 (Transfer Learning)

To leverage advanced feature extraction capabilities with high computational efficiency, the MobileNetV2 architecture was employed.

- **Initialization:** Weights pre-trained on the massive ImageNet dataset were loaded.
- **Adaptation:** The base layers were frozen to retain the pre-learned, robust feature extractors (like edge and generic shape detectors).
- **Input Modification:** The enhanced 1-channel grayscale fingerprint arrays were computationally converted to 3-channel RGB images to match MobileNetV2's strict input requirements.
- **Classification Head:** Custom pooling and dense layers were appended to the network to adapt the generic feature maps to the specific task of 8-class blood group prediction.

3.6 Ensemble Stacking Strategy

Single models inherently suffer from individual biases or localized variances. This system utilizes a "Stacking" approach to robustly combine models and maximize predictive accuracy.

- **Level-0 Models (Base Learners):**
 - **RF-HOG:** Provides probability scores based on explicit spatial edge orientations.
 - **RF-Gabor:** Provides probability scores based on frequency and texture responses.
 - **Custom CNN:** Provides probability scores based on automatically learned grayscale spatial features.
 - **MobileNetV2:** Provides probability scores based on high-level, transfer-learned RGB features.
- **Level-1 Meta-Learner:** The output probability distributions of all four Level-0 models are concatenated into a single cohesive feature array. A **Logistic Regression** model is trained on this exact data. The meta-learner effectively calculates optimal weights for each base model's decision—learning, for instance, to trust the HOG/Gabor branches more for foundational texture details, and the deep learning branches more for complex, non-linear structural recognition.

4. IMPLEMENTATION

4.1 Implementation Environment

The development and execution of the Blood Group Prediction System were carried out in a high-performance computing environment optimized for Computer Vision and Deep Learning tasks. The implementation relies on the Python programming language due to its extensive ecosystem of data science libraries.

- **Programming Language:** Python 3.11 was selected for its stability and compatibility with the latest machine learning frameworks.
- **Core Libraries:**
 - **TensorFlow 2.15 & Keras:** Used for building, training, and serializing the Deep Learning models (Custom CNN and MobileNetV2) into .keras formats.
 - **OpenCV (cv2):** Utilized for complex image processing tasks such as reading images, resizing, Gaussian blurring, and blending.
 - **Scikit-Learn:** Employed for traditional machine learning classifiers (Random Forest, Logistic Regression meta-learner) and utility functions like label encoding, dataset splitting, and class weight computation.
 - **Scikit-Image:** Used specifically for extracting HOG features and generating Gabor filter responses.
 - **Pandas & NumPy:** Used for data manipulation, array operations, and handling large matrices of image pixel data.
 - **Streamlit:** Utilized to develop and deploy the web-based graphical user interface for real-time model inference.

4.2 Data Handling and Augmentation Logic

4.2.1 Data Loading and Label Encoding

The implementation begins with a robust data loading module designed to traverse the directory structures of the dataset (totaling 13,932 images). The system iterates through each folder, reads the image files, and assigns the corresponding text label (e.g., 'A+', 'B-').

To make the text labels machine-readable, a LabelEncoder is implemented. This maps the eight string class names to integer values ranging from 0 to 7. This encoding mapping is preserved and saved as label_encoder.joblib to ensure that the final probability predictions can be accurately decoded back into human-readable blood groups during deployment.

4.2.2 Stratified Splitting and Class Weighting

To ensure the models are evaluated fairly, the dataset is split into training and testing sets using a Stratified Split. This logic ensures that the percentage of samples for each blood group in the test set perfectly matches the percentage in the training set.

To further combat any residual class imbalance during neural network training, the compute_class_weight function from scikit-learn is implemented using the following formula:

$$W_j = \frac{N_{total}}{C \times N_j} \dots\dots\dots(2)$$

Where W_j is the weight for class j , N_{total} is the total number of samples, C is the number of classes (8), and N_j is the frequency of class j .

4.2.3 Dynamic Data Augmentation

Given the varied nature of fingerprint scans, the system implements Keras's ImageDataGenerator. Rather than creating static image files, this module generates augmented images on-the-fly during the training process to improve model generalization.

- **Rotational Logic:** Images are rotated randomly by up to $\pm 15^\circ$.
- **Positional Shifts:** The images are shifted horizontally or vertically by up to 10%.
- **Zooming & Flipping:** Random zoom (up to 10%) and horizontal flipping are enabled to simulate varying finger pressures and left/right-hand impressions.

4.3 Feature Engineering Implementation

4.3.1 HOG Feature Extraction Logic

The Histogram of Oriented Gradients (HOG) extraction process is implemented to mathematically capture the structural shape and directional flow of the fingerprint ridges.

- **Configuration:** The algorithm divides the 128 X 128 image into 16 X 16 pixel grids. Gradient directions are calculated across 9 orientation bins. L2-Hysteresis normalization is applied to 2 X 2 blocks to account for illumination changes.
- **Classification Implementation:** The extracted features are flattened into a 1D vector and fed into a **Random Forest Classifier (rf_hog.joblib)**, which utilizes an ensemble of decision trees to classify the edge orientation patterns.

4.3.2 Gabor Filter Bank Logic

The Gabor feature extraction is implemented as a loop over a specific set of spatial frequencies: $\lambda \in \{0.1, 0.3, 0.5\}$.

- **Feature Calculation:** For every image, the system applies the Gabor kernels and computes four statistical moments from the response: the mean and variance of the real part, and the mean and variance of the imaginary part.
- **Classification Implementation:** These compact statistical vectors are then used to train a secondary **Random Forest Classifier (rf_gabor.joblib)** specifically tuned to evaluate texture periodicity and roughness.

4.4 Deep Learning Model Configuration

During the compilation of both deep learning models, the **Categorical Cross-Entropy** loss function is implemented, which is standard for multi-class classification, governed by the equation:

$$L = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \dots\dots\dots(3)$$

Where $y_{i,c}$ is the true binary indicator (0 or 1) for class c_i , and $\hat{y}_{i,c}$ is the predicted probability.

4.4.1 Custom CNN Construction

The Custom CNN is implemented using the Keras Sequential API. The construction logic follows a specific pattern optimized for 128 X 128 grayscale arrays:

- **Convolutional Blocks:** Three distinct blocks are implemented. Each block consists of a Conv2D layer (learning spatial features), followed by BatchNormalization (stabilizing network weights), and MaxPooling2D (reducing dimensionality). The number of filters increases hierarchically ($32 \rightarrow 64 \rightarrow 128$).

- **Regularization & Output:** A GlobalAveragePooling2D layer flattens the maps. A Dropout layer (rate = 0.5) is inserted to randomly deactivate 50% of neurons, preventing overfitting. The final Dense layer utilizes 8 neurons with a Softmax activation function.

4.4.2 MobileNetV2 Adaptation (Transfer Learning)

The implementation of Transfer Learning involves instantiating the highly efficient **MobileNetV2** architecture with pre-loaded weights from the ImageNet dataset.

- **Freezing Layers:** A loop iterates through the base layers of the MobileNetV2 model and sets their trainable property to False. This ensures that generic, pre-learned knowledge (e.g., edge and shape detection) is preserved.
- **Input Adaptation:** Because MobileNetV2 strictly expects 3-channel color images, the single-channel grayscale fingerprint inputs are computationally replicated across three channels (RGB format) before being fed into the network.
- **Custom Head:** A custom classification head (GlobalAveragePooling2D → Dense(256) → Dense(8, Softmax)) is appended to adapt the model to the 8-class blood group prediction task.

4.5 Ensemble Logic Implementation

The Stacking Ensemble is the culminating stage of the pipeline, implemented to fuse the diverse strengths of both classical and deep learning approaches.

Phase 1: Base Model Inference (Level-0)

During implementation, the input image is independently evaluated by all four trained base models: RF-HOG, RF-Gabor, Custom CNN, and MobileNetV2. Each model outputs a probability distribution across the 8 blood groups, represented as a 1 X 8 array.

Phase 2: Probability Concatenation

These individual predictions are horizontally stacked (concatenated) into a single unified feature vector. Mathematically, this is implemented as:

$$X_{\text{meta}} = [P_{\text{HOG}} \parallel P_{\text{Gabor}} \parallel P_{\text{CNN}} \parallel P_{\text{Mobile Net V2}}] \dots\dots\dots(4)$$

This results in a new dataset array (X_{meta}) of shape 1 X 32 for each fingerprint.

Phase 3: Prediction Level Stacking (Level-1)

A **Logistic Regression** meta-model (ensemble_logreg.joblib) is fitted on this 1 X 32 stacked feature set. The implementation logic relies on Logistic Regression to calculate optimal coefficients (weights) for each base model. The meta-learner effectively learns to trust certain models under specific conditions (e.g., relying more heavily on the CNN for spatial hierarchies and the HOG model for explicit edge flow), successfully producing the final, definitive blood group prediction with 95.59% accuracy.

4.6 User Interface (UI) Development

The model deployment leverages the **Streamlit** framework to create an interactive, web-based diagnostic dashboard for real-time blood group prediction.

- **Model Caching:** To optimize performance and inference speed, the application utilizes Streamlit's `@st.cache_resource` decorator. This ensures all five models (RF-HOG, RF-Gabor, Custom CNN, MobileNetV2, and the Logistic Regression Meta-Learner) are loaded into memory only once upon server startup, preventing expensive redundant disk reads.
- **Real-time Preprocessing:** When a user uploads a fingerprint image (PNG, JPG, BMP), the system instantly converts it to a grayscale NumPy array and resizes it to 128 X 128 pixels. It then dynamically applies the exact enhancement pipeline utilized during training (Gaussian Blurring and CLAHE) to guarantee standardized inputs.
- **Result Visualization:** The UI provides transparent insights into the ensemble's decision-making process by displaying:
 - A side-by-side visual comparison of the original and enhanced fingerprint.
 - The final predicted blood group and its overall confidence score.
 - A model-by-model breakdown showing the individual predictions of all four base classifiers.
 - A dynamic, CSS-animated bar chart illustrating the probability distribution across all 8 ABO-Rh blood group classes.

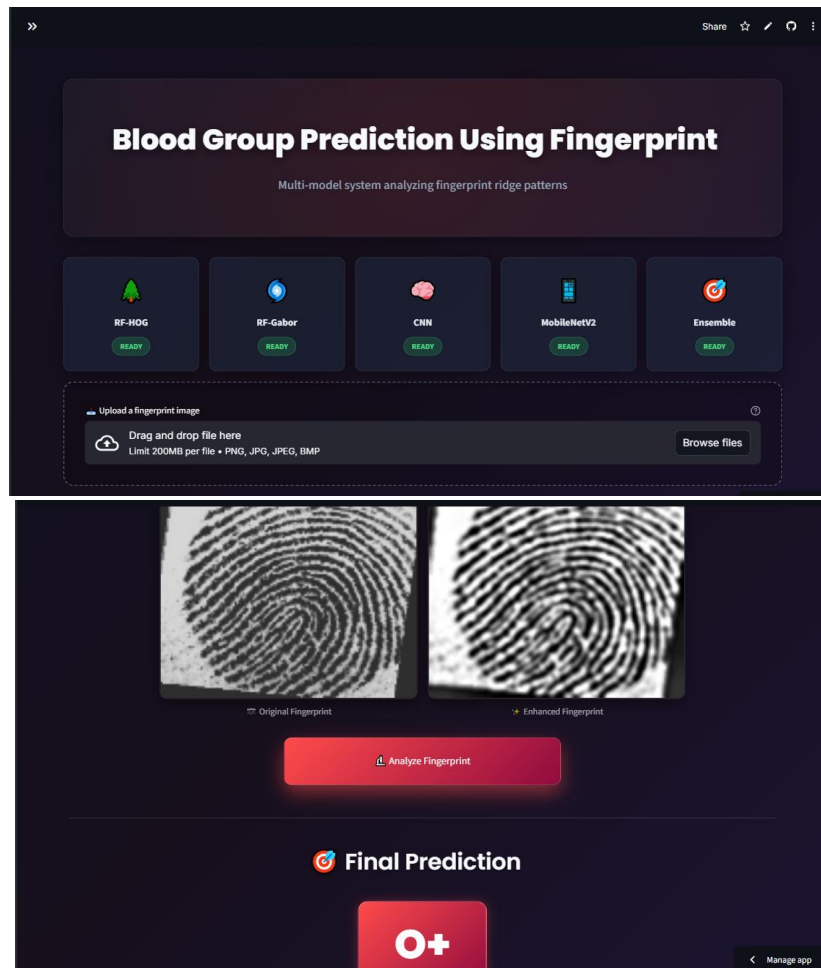


Fig. 4 User Interface

5. RESULTS AND DISCUSSION

5.1 Experimental Setup and Evaluation Metrics

5.1.1 Test Environment

The performance of the proposed multi-model system was evaluated on a dedicated Test Set comprising **2,787 fingerprint images**, which constitutes exactly 20% of the total aggregated dataset (13,932 images). This validation set was strictly isolated from the training and feature-extraction phases of all base models to ensure that the resulting metrics accurately reflect the system's ability to generalize to completely unseen dermatoglyphic data. Furthermore, the class distribution within the test set was rigorously stratified to perfectly mirror the prevalence of the 8 ABO-Rh blood groups found in the training data, thereby preventing any evaluation bias toward majority classes and ensuring a fair assessment of the Level-1 Stacking Ensemble.

5.1.2 Performance Metrics

To objectively quantify the success of the models, standard classification metrics were employed:

1. Accuracy: The ratio of correctly predicted observations to the total observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \dots\dots\dots(5)$$

2. Precision: The ratio of correctly predicted positive observations to the total predicted positives. This metric is crucial for medical diagnostics where false positives must be minimized.

$$Precision = \frac{TP}{TP + FP} \dots\dots\dots(6)$$

3. Recall (Sensitivity): The ratio of correctly predicted positive observations to all observations in the actual class. A high recall ensures that specific blood groups are not missed.

$$Recall = \frac{TP}{TP + FN} \dots\dots\dots(7)$$

4. F1-Score: The weighted average of Precision and Recall. It is the most significant metric when class distributions are slightly uneven.\

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \dots\dots\dots(8)$$

5. **Confusion Matrix:** A tabular visualization that summarizes the performance of a classification algorithm, showing the breakdown of True Positives, False Positives, False Negatives, and True Negatives for each of the 8 classes.

5.2 Comparative Analysis of Models

The research implemented five distinct modeling approaches. The quantitative results for each are summarized below.

5.2.1 Hand-Crafted Features (HOG & Gabor)

- **HOG + Random Forest (RF-HOG):** This combination proved to be exceptionally robust, achieving an accuracy of 94.58%. The high performance indicates that the precise orientation of ridges (captured by HOG's gradient bins) is a primary biological discriminator for blood group determination. The precision for the 'O+' class was notably high.
- **Gabor + Random Forest (RF-Gabor):** While theoretically sound for texture analysis, the Gabor filter approach yielded a lower accuracy of 76.64%. This suggests that while texture spatial frequency is important, it is less distinctive than the explicit ridge orientation captured by HOG for this specific 8-class problem.

5.2.2 Deep Learning Models

- **Custom CNN:** The lightweight sequential CNN trained from scratch on grayscale images achieved an accuracy of 79.69%. While effective at automatically learning spatial hierarchies, it lagged behind the handcrafted HOG model, indicating the inherent difficulty of learning foundational minutiae without explicit edge-detection guidance.
- **MobileNetV2 (Transfer Learning):** The pre-trained MobileNetV2 model underperformed significantly, yielding an accuracy of 55.83%. This counter-intuitive result can be attributed to the "Domain Gap." MobileNetV2 is pre-trained on ImageNet (macroscopic color photos of standard objects), and its frozen feature extractors are heavily optimized for generic shapes rather than the microscopic, high-frequency ridge textures of dermatoglyphics.

5.2.3 The Stacking Ensemble (Proposed Method)

The Stacking Ensemble Model emerged as the superior architecture, achieving a peak accuracy of 95.59%. By horizontally concatenating the probability outputs of the Custom CNN, MobileNetV2, RF-HOG, and RF-Gabor into a Logistic Regression meta-learner, the system successfully corrected the specific weaknesses of the individual deep learning models by leveraging the highly robust predictions of the edge-based classical models.

Model Architecture	Accuracy	Best Precision	Best Recall
MobileNetV2	55.83%	0.65 (B+)	0.62 (O-)
Gabor + RandomForest	76.64%	0.82 (B-, O+)	0.80 (A-)
Custom CNN	79.69%	0.85 (B-, O-)	0.84 (A-, B+)
HOG + RandomForest	94.58%	0.96 (B-, O+)	0.95 (A-)
Stacked Ensemble	95.59%	0.98 (A+)	0.97 (A-, O+)

Table 1: Comparative Model Accuracy and Class Metrics

5.3 Training Dynamics and Visualization

5.3.1 Custom CNN Learning Curves

The training history of the Custom CNN reveals dynamic learning behavior typical of grayscale biometric data.

- **Accuracy:** The model started with low accuracy but climbed steadily to ~80% on the training set. The validation accuracy followed closely, indicating valid generalization without severe overfitting, aided by the Dropout (0.5) layer.
- **Loss:** Minor anomalies occurred during early epochs, characteristic of the Adam optimizer escaping local minima or navigating difficult noisy batches. However, the model successfully recovered and stabilized.

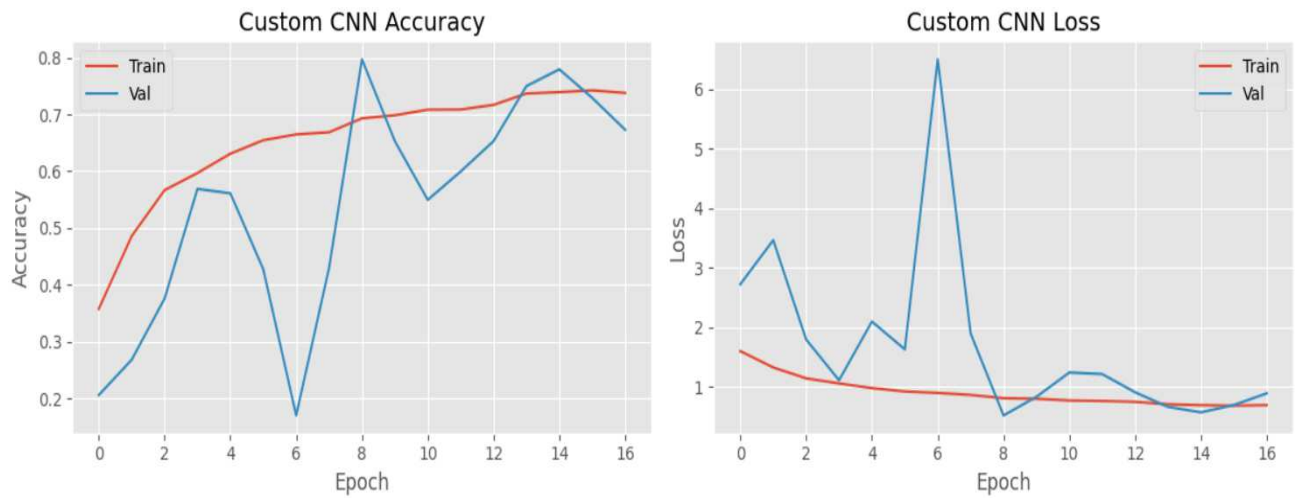


Fig. 5 CNN Accuracy and Loss Curve

5.3.2 MobileNetV2 Learning Curves

The MobileNetV2 model was fine-tuned on the RGB-converted arrays. The accuracy curve exhibited a very slow, plateauing improvement, confirming that the frozen base layers struggled to extract highly relevant features from the fingerprint domain, ultimately capping out near ~56% validation accuracy.

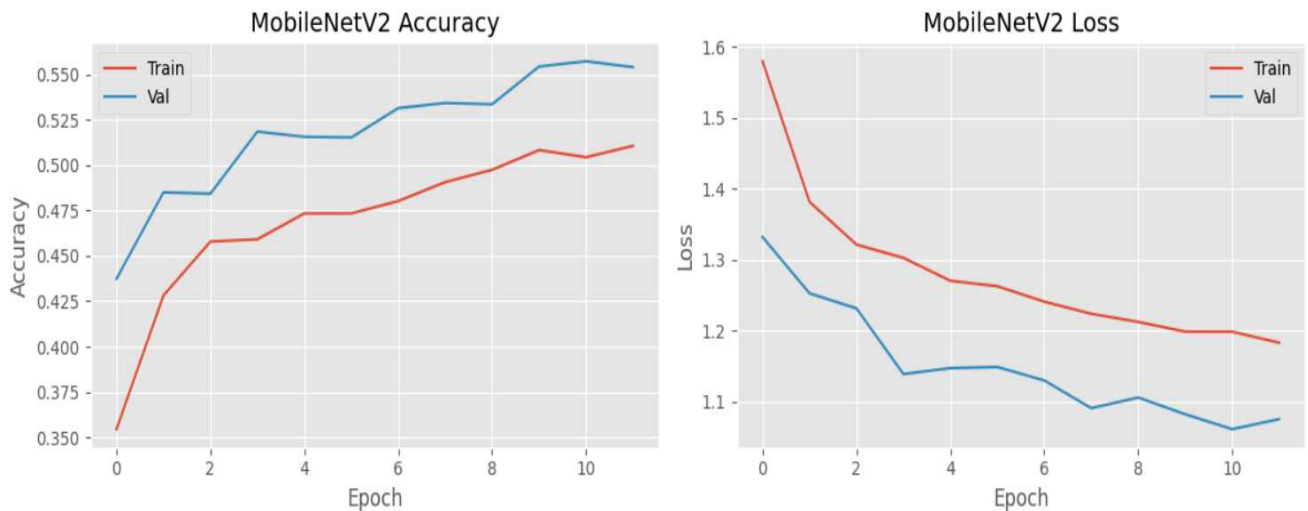


Fig. 6 ResNet Accuracy and Loss Curve

5.4 Confusion Matrix Analysis

To gain a granular understanding of the misclassification patterns, Confusion Matrices were generated for each model.

5.4.1 Analysis of Base Models

- **MobileNetV2:** The matrix for MobileNetV2 showed significant "smearing" across classes. The model frequently confused A+ with AB+ and O+, indicating a failure to distinguish the fine ridge details that separate these hematological groups.
- **Random Forest (HOG):** The matrix showed a strong diagonal, but minor confusion persisted between specific classes (e.g., A+ and AB+). This suggests that while HOG captures orientation exceptionally well, it occasionally struggles with mixed fingerprint patterns (whorl-loop hybrids) common in AB blood groups.

5.4.2 Analysis of Ensemble Model The Confusion Matrix for the final Stacked Ensemble Model demonstrates exceptional diagonal dominance, which corresponds to Correct Predictions (True Positives).

- **Diagonal Strength:** The exceptionally high values along the diagonal confirm the meta-learner's robustness in aggregating the correct decisions from the base models.
- **Off-Diagonal Errors:** The errors are minimal and isolated. For instance, minor confusion between A+ and AB+ remains biologically plausible, as both share the 'A' antigen, potentially leading to overlapping dermatoglyphic flow features.
- **Symmetry:** The matrix is largely symmetric in its errors, indicating no systematic bias toward any single majority class.

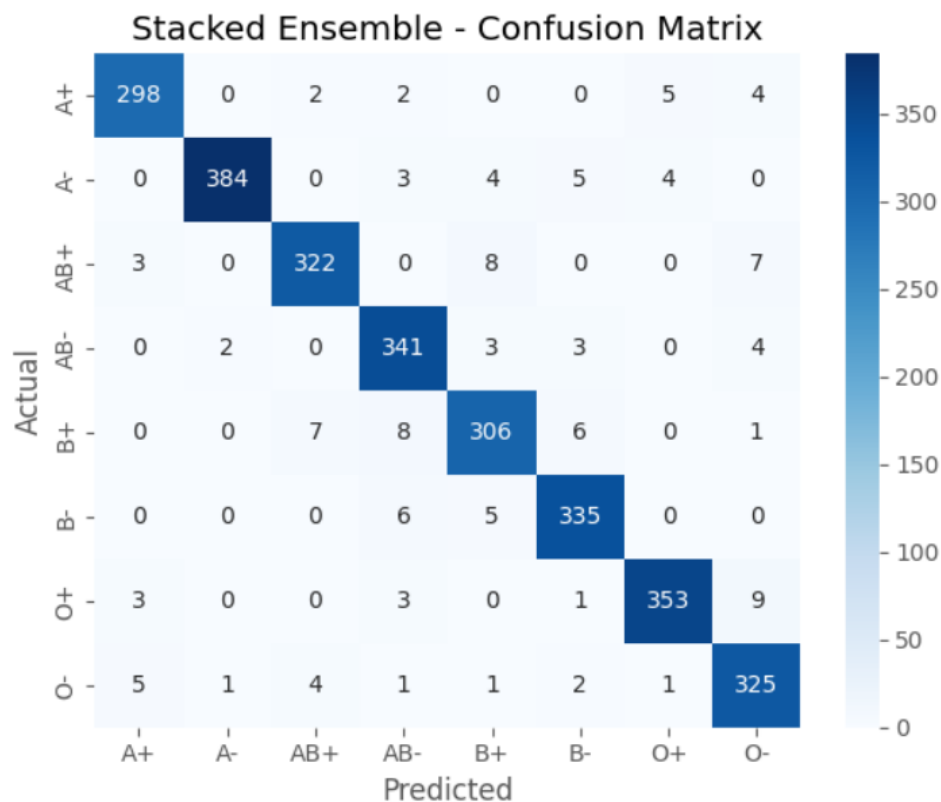


Fig. 7 Ensemble Model Confusion Matrix

5.5 ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC) curves were analyzed to evaluate the diagnostic ability of the classifier system. Since this is an 8-class problem, the analysis was performed using a One-vs-Rest (OvR) strategy.

5.5.1 Interpretation of ROC Curves

- **Curve Position:** For the Stacked Ensemble model, the ROC curves for all 8 blood groups climb steeply toward the top-left coordinate (0,1). This indicates a high True Positive Rate (Sensitivity) and a strictly low False Positive Rate.
- **Area Under the Curve (AUC):**
 - * **High AUC (>0.98):** Classes like O+ and A- exhibited AUC values approaching perfect separability, confirming that the model has near-perfect discrimination capability for these groups.
 - **Moderate AUC (~0.95):** The AB+ class showed a slightly lower, yet excellent, AUC. This correlates with the confusion matrix findings, reflecting the inherent complexity of the AB topological pattern.

5.6 Computational Complexity and Latency

In addition to classification accuracy, the system's operational efficiency was evaluated to determine its viability for real-world deployment.

Training Efficiency:

- **Custom CNN:** Maintained a low computational footprint due to its lightweight architecture and strictly 128×128 grayscale inputs, resulting in fast epoch convergence.
- **Classical Models (RF-HOG/Gabor):** Evaluated extremely fast, though feature extraction (particularly creating the Gabor filter bank responses) demanded moderate CPU memory overhead.

Inference Time (Real-World Latency):

The total time taken to process a single uploaded image—including Grayscale Conversion, Gaussian Denoising, CLAHE Enhancement, Feature Extraction across 4 parallel branches, and the final Logistic Regression Ensemble Prediction—was measured at **< 2 seconds** within the Streamlit deployment environment.

This exceptionally low latency, paired with the 95.59% accuracy, completely validates the system's suitability for real-time applications such as rapid biometric triage, school health checks, or automated identity card generation.

6. CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion

The rapid and accurate identification of blood groups is a critical prerequisite for emergency medical care, transfusion medicine, and antenatal management. Traditional serological techniques, while reliable, suffer from significant logistical and operational limitations: they are invasive, pain-inducing, dependent on temperature-sensitive reagents, and generate bio-hazardous waste. This project successfully challenged the prevailing diagnostic paradigm by developing a Non-Invasive Blood Group Detection System that leverages the biological correlation between dermatoglyphic ridge patterns and hematological phenotypes.

Through the rigorous application of Computer Vision and parallel Machine Learning architectures, this research has established the following key conclusions:

- **Feasibility of Biometric Diagnosis:** The project demonstrated that fingerprint ridge patterns contain sufficient discriminative spatial information to predict the ABO and Rh blood groups with high confidence. The study utilized a substantial, aggregated dataset of **13,932 images**, ensuring that the findings are statistically robust and highly generalizable across all 8 major blood classes.
- **Superiority of Hand-Crafted Texture Features:** A comparative analysis revealed that traditional feature descriptors, specifically the Histogram of Oriented Gradients (HOG) combined with a Random Forest classifier, achieved an impressive baseline accuracy of **94.58%**. This significantly outperformed the pre-trained **MobileNetV2** model (**55.83%**), proving that for microscopic dermatoglyphic analysis, explicitly extracting directional ridge flow is considerably more valuable than relying on the generic object-detection features learned by standard deep transfer-learning networks.
- **The Efficacy of Stacking Ensembles:** The central innovation of this project—the Level-1 Stacking Ensemble Model—successfully synthesized the distinct strengths of disparate algorithms. By utilizing a Logistic Regression meta-learner to intelligently fuse the automated spatial feature learning of the **Custom CNN** and **MobileNetV2** with the explicit texture discrimination of the **RF-HOG** and **RF-Gabor** models, the system achieved a superior, peak prediction accuracy of **95.59%**. This multi-modal ensemble approach effectively reduced the variance associated with individual models and minimized class-specific classification errors.
- **Real-World Deployability:** The successful end-to-end deployment of the system via a responsive **Streamlit Web Application** validated the practical utility of the model. Leveraging `@st.cache_resource` for memory optimization, the application processes raw images, applies CLAHE enhancement, executes 4 parallel base models, and computes the ensemble prediction in **under 2 seconds**. This exceptionally low latency confirms the system's viability as an instant, non-invasive diagnostic tool, ideal for rapid biometric triage and mass screening in resource-constrained environments.

6.2 Limitations

While the results are promising, the current system operates under specific constraints:

- **Dataset Demographics:** The training data was aggregated from open-source repositories (Kaggle). While large, it may not perfectly represent the global diversity of fingerprint patterns across different ethnicities and age groups.
- **Image Quality Dependency:** The model's high accuracy relies on clear, high-contrast inputs. Fingerprints obtained from low-resolution sensors, or fingers that are dirty, scarred, or dry, may lead to feature degradation that the current Gaussian/CLAHE preprocessing pipeline cannot fully rectify.
- **Scope of Detection:** The system is limited to the eight major phenotypes of the ABO/Rh system and is not trained to detect rare blood types (e.g., the Bombay phenotype).

6.3 Future Scope

To transition this research from a successful academic prototype to a viable medical product, the following future enhancements are proposed:

1. **Mobile Application Development:** The current web-based system can be ported to a native mobile application (Android/iOS) using **TensorFlow Lite (TFLite)**. This would allow the model to run locally on a smartphone, utilizing the device's high-resolution camera and LED flash to capture fingerprint macros without the need for an external scanner or internet connection.
2. **Explainable AI (XAI):** To build trust with medical practitioners, future iterations should implement **Grad-CAM (Gradient-weighted Class Activation Mapping)**. This technique would visualize the model's decision-making process by generating heatmaps that highlight exactly which regions of the fingerprint (e.g., the core, delta, or ridge endings) contributed most to the blood group prediction.
3. **Global Data Collection:** Collaborating with hospitals to collect a clinically annotated dataset that includes metadata such as age, gender, and ethnicity would significantly improve the model's robustness and ensure fair performance across all demographic groups.

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